

VANSCANTM

Typhoid RT-PCR



Typhoid RT-PCR test

First of its kind **Patented* RT-PCR Test** for the detection of Salmonella typhi & paratyphi In human blood (Patent No. 355208)

PATENTED TECHNOLOGY



VANSCANTM

Facts & Figures

Causative organisms : Salmonella typhi & Salmonella paratyphi

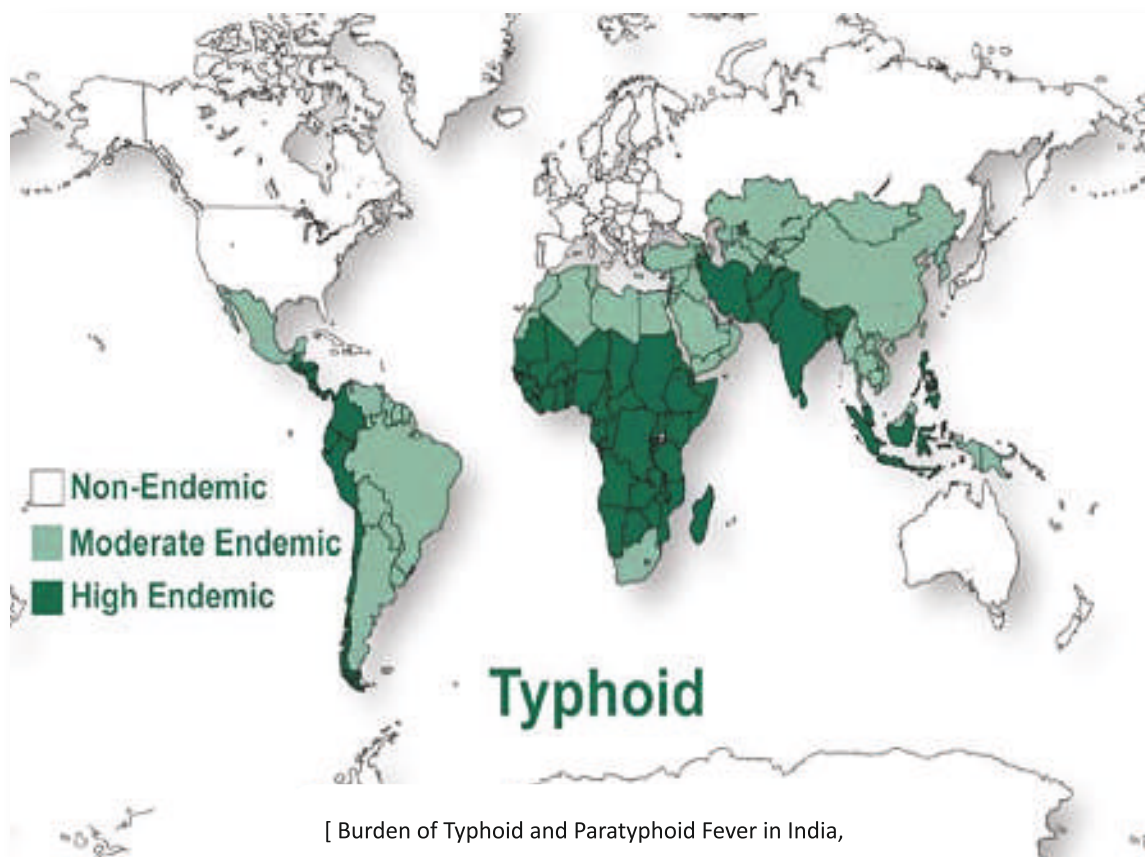
No. of cases /year globally : > 14 Million ~ Half in India

Route of infection : Fecal - Oro route

Early & Accurate Diagnosis

Imperative to initiate the right treatment

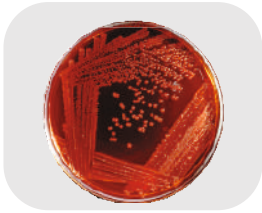
To reduce the risk of Antimicrobial Resistance (AMR)



HIGH ENDEMICITY IN INDIA

VANSCANTM

Lab Diagnosis of Typhoid - a Challenge

Type of Typhoid Detection Test	Performance Characteristics	
Widal Tube & Slide Test	Obsolete, Unstandardised	
Rapid Card Tests	Unreliable, False +ve / False - ve results	
Conventional Blood Culture	1. Confirmatory test takes > 5days 2. Results affected by intake of antibiotics 3. Needs multiple blood samples 4. Cumbersome	
VANSCAN TM Typhoid RT-PCR	1. The New Gold Standard 2. Reliable & Rapid 3. Compatible with open RT-PCR Systems	

FULFILLING THE UNMET NEED

VANSCANTM

Typhoid RT-PCR Test



- RT-PCR based Assay
- Patented Primers & Probes Sequences
- Targeting Unique Regions of Salmonella typhi & paratyphi
- Highly Sensitive & Specific

An Industry - Academia Collaboration with
the Graphic Era (Deemed to be University) Dehradun.

PATENTED RT-PCR TEST FOR TYPHOID



VANSCANTM

Typhoid RT-PCR Test

Multi Centric Validation

- Out of the 153 culture negative samples tested, 5 samples were found to be positive with VANSCAN Typhoid RT PCR Test.
- VANSCANTM Typhoid RT PCR Test was more sensitive than the culture method.

Validation Site	Culture vs VANSCAN Typhoid RT PCR Test		Culture vs VANSCAN Typhoid RT PCR Test	
	Culture Positive	RT PCR Positive	Culture Negative	RT PCR Negative
Regional Centre for Bio- technology (DBT-Lab), Faridabad	15	15	7	7
Graphic Era University, Dehradun	84	84	*153	148
Validation Site	Reference Kit vs RT PCR Positive		Reference Kit vs RT PCR Negative	
NABL Accredited Lab, Jaipur	8	8	14	14
Total	107	107	174	169

SENSITIVE, SPECIFIC, RELIABLE

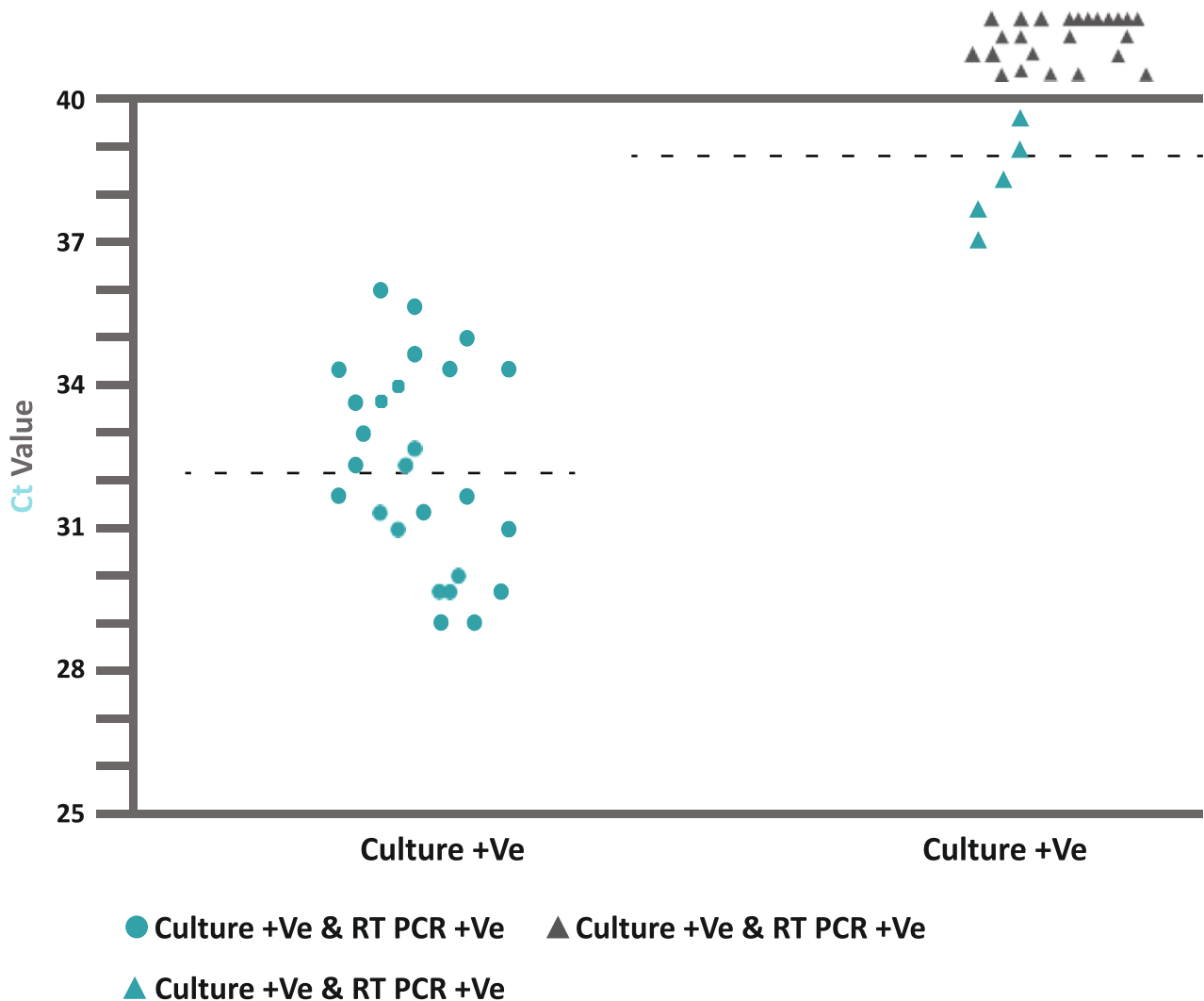


VANSCANTM

Typhoid RT-PCR Test

Comparison with Blood Culture

Culture vs VANSCAN RT PCR Performance



HIGHLY SENSITIVE TEST



Typhoid RT-PCR Test

Comparative Analysis - RT-PCR Test for Typhoid Detection

Parameter	VANSCAN Typhoid RT PCR Test	Liferiver China	DiagSure Typhoid	True NAT Salmonella
What does the test detect?	Salmonella typhi & para typhi	Salmonella typhi	Salmonella typhi	Salmonella typhi & para typhi
Does the test require a culture step?	No Extracted DNA can be tested directly	Pre-extraction step is required	No Extracted DNA can be tested directly	Pre- test Culture is mandatory
Compatibility with RT PCR Systems	Can be used with any open system	Can be used with any open system	Can be used with any open system	Is a closed system
Time required for the test	~ 1 hour	~ 1 hour	1 hour 5 mins	5 to 24 hours
Analytical Sensitivity	10 – 100 CFU / ml	1 X 10 ³ CFU / ml	>1000 CFU / ml	327 – 716 CFU / ml
Storage Temperature	-20°C	-20°C	-20°C	2 - 30°C
Gene detected	- Outer Membrane Protein- FAM - RNAse P – HEX	-	StaG gene	‘pmp’ gene

THE NEW GOLD STANDARD



